Algorithms and Data Structures 1

Chapter 1: Introduction to Computer Science

Exercise Series (30 Questions) and Complete Answer Key

This exercise series covers the concepts studied in Chapter 1: the notion of computer science, the structure and operation of a computer, and the concept of an algorithm. It includes 30 questions (the 17 original questions from the course handout, supplemented by 13 new questions), followed in the second part of the document by the complete answer key for each question.

Part 1 - Exercise Series

Question 1. What is computer science? What is a computer? Compare a human being and a computer.

Question 2. What is information? What is the difference between data and information?

Question 3. What are the differences between machine language, assembly language, and high-level language?

Question 4. What is a computer system?

Question 5. What are the components of a computer (PC architecture)?

Question 6. What is the purpose of a motherboard?

Question 7. In computer equipment, data is represented by an electrical signal. What form does this signal take?

Question 8. Identify the following devices and state whether they are input, output, or input/output devices: Keyboard, Monitor, Printer, DVD drive, Scanner, Webcam, DVD writer, Mouse, Floppy-disk drive, CD-ROM drive, Microphone, Headset, Speakers, CD-ROM writer, USB connector.

Question 9. In the following table, indicate the type of operation (Input, Output, Storage) and the device used:

Operation	Type of operation	Device
I type a text		
I scan a graph		
I print results		
I save data		
I listen to music from my PC		

Question 10. What does main memory contain?

Question 11. What is the difference between RAM and ROM?

Question 12. What is the difference between main memory and auxiliary memory?

Question 13. Software can be classified in several ways. Explain and describe the classification of software according to its importance.

Question 14. Give a simplified functional diagram of a computer.

Question 15. Which characteristics can be used to assess a computer's performance? (Only mention those related to the processor, memory, and storage units.)

Question 16. I am in the street and I want to go home (to an apartment building where I must take the elevator). Propose an algorithm.

Question 17. We want to solve in R the equation $ax^2 + bx + c = 0$; that is, for the real data a, b, and c, give the real solution or solutions of the equation. Propose an algorithm.

Additional Questions (18 to 30)

Question 18. Convert 2.5 GB (gigabytes) into MB, then KB, then bytes.

Question 19. Knowing that the decimal ASCII code of the character 'a' is 97, give its binary equivalent, i.e. ASCII('a').

Question 20. Explain the role of the data bus in the operation of a computer.

Question 21. What is the difference between a computer program and software?

Question 22. Recall the three types of software according to the services they provide to users, and give an example of each.

Question 23. Where does the word "Algorithm" come from?

Question 24. List and briefly explain the five properties that an algorithm must satisfy.

Question 25. For each of the following situations, indicate which property of an algorithm is not respected:

- a) An algorithm that, for certain input data, never stops.
- b) An algorithm containing the instruction "divide approximately by a small number".
- c) An algorithm designed to solve only one particular case of a problem, without considering the other possible cases.
- d) An algorithm that solves a problem correctly but consumes excessive time and resources even though a faster solution exists.

Question 26. What is a pseudo-algorithm? Give an example.

Question 27. Propose an algorithm that calculates and displays the sum of two integers entered by the user.

Question 28. Propose an algorithm that determines and displays the largest of three real numbers.

Question 29. Propose a pseudo-algorithm (in natural language) describing the steps for preparing a cup of coffee or tea.

Question 30. A computer has 4 GB of RAM. Knowing that each ASCII character occupies 1 byte (8 bits), how many characters can be stored at most in this memory?

Part 2 - Answer Key

Answer 1. Definitions and comparison.

Computer science is the automatic (logical and orderly) processing of information, essentially using a machine called a computer. The word is a portmanteau formed from "information" and "automatic."

A computer is a very fast electronic machine (calculator) that accepts digital information as Input, processes it according to the program stored in its Main Memory, and produces information (the result) as Output.

Human being / computer comparison: both follow the same general Input -> Processing -> Output pattern. In humans, the senses (eyes, ears, etc.) play the role of input units, the brain that of the central processing unit (memory and reasoning), and speech or gestures that of output units. The essential difference is that human beings exercise judgment, creativity, and autonomous learning, whereas a computer simply executes, very quickly and without error, precise instructions defined in advance by a program, without understanding or awareness of what it is doing.

Answer 2.

Information is data that has meaning: Information = (data, meaning).

Data is represented by one or more characters (alphabet, punctuation marks, special symbols, etc.) and does not necessarily have meaning.

Difference: all information is data, but not all data is necessarily information. Example: "Xwil3?#" is meaningless data, whereas "keyboard" is information because it has meaning (an image can even be associated with it).

Answer 3. Machine language, assembly language, high-level language.

- Machine language: a sequence of 0s and 1s directly executable by the processor; it depends closely on the processor type and is not readable by a human.
- Assembly language: a symbolic (mnemonic) representation of machine language, more readable for humans, but still specific to a processor type; it must be translated into machine language by an assembler.
- High-level language: a language close to natural language (e.g., C, Python, Java), machine-independent, which must be translated into machine language by a compiler or an interpreter.

Answer 4.

A computer system is the organized set of hardware resources, software resources, data, and personnel used to acquire, process, store, and communicate information within an organization.

Answer 5. Components of a computer (PC architecture):

- The motherboard;
- The processor (CPU);
- Main memory (RAM, ROM);
- Storage memory (hard disk, SSD, etc.);
- The graphics card (and possibly a sound card and network card);
- The power supply unit;
- Input devices (keyboard, mouse, scanner, etc.) and output devices (monitor, printer, etc.);
- Buses (data, address, control) and input/output ports (USB, HDMI, etc.).

Answer 6.

The motherboard is the computer's main printed circuit board: it connects all components (processor, memory, expansion cards, peripherals) and enables communication between them through buses, and it distributes electrical power to these components.

Answer 7.

Data is represented by a binary (digital) electrical signal, that is, a square-wave signal that alternates between two voltage levels corresponding to the values 0 and 1.

Answer 8. Device classification:

Device	Type
Keyboard	Input
Monitor	Output
Printer	Output
DVD drive	Input
Scanner	Input
Webcam	Input
DVD writer	Input/Output
Mouse	Input
Floppy-disk drive	Input/Output
CD-ROM drive	Input
Microphone	Input
Headset	Output
Speakers	Output
CD-ROM writer	Input/Output
USB connector	Input/Output

Note: a reader/drive (DVD, CD-ROM) only reads a medium and is therefore an input device; a writer can read from and write to the medium, so it is an input/output device. The USB connector is a port that can accommodate both input and output devices.

Answer 9. Completed table:

Operation	Type of operation	Device
I type a text	Input	Keyboard
I scan a graph	Input	Scanner
I print results	Output	Printer
I save data	Storage	Hard disk / secondary memory (USB flash drive, etc.)
I listen to music from my PC	Output	Loudspeakers / speakers (or headset)

Answer 10.

Main memory (central memory) contains: the operating system (the part currently loaded and running), the programs currently running, and the data currently being processed by the processor.

Answer 11. RAM and ROM:

- RAM (Random Access Memory): volatile memory (loses its contents when electrical power is absent), readable and writable, and stores information currently being processed.
- ROM (Read Only Memory): non-volatile memory, generally read-only; it contains, for example, the BIOS (Basic Input Output System), a program supplied by the machine manufacturer.

Answer 12. Main memory vs. auxiliary memory:

- Main memory (MM): very fast access, generally volatile, low capacity; it is essential because data can be processed only if it is in main memory.
- Auxiliary / secondary memory (hard disk, CD, DVD, etc.): slower access, non-volatile, large capacity; it is used for long-term storage of information.

Answer 13. Classification of software according to importance:

- Basic (or system) software: essential to the operation of the machine itself; it acts as an intermediary between the hardware and the user (or other software). Examples: operating system, BIOS, drivers.
- Application software: runs on top of the operating system and meets specific user needs. It includes standard software intended for a broad public (word processing, spreadsheets, etc.) and software packages developed for a particular need (payroll, accounting, stock management, etc.).

Answer 14. Simplified functional diagram of a computer:

Input Unit (IU)	-> BUS ->	**Central Processing Unit (CPU)**	-> BUS ->	**Output Unit (OU)**
Examples: keyboard, mouse, scanner... (UNPROCESSED information)		• Main Memory (MM) • Arithmetic and Logic Unit (ALU) • Control Unit (CU)		Examples: monitor, printer... (PROCESSED information)

Unprocessed information enters through the IU and travels via the data bus to main memory; the ALU processes it under the control of the CU (with exchanges involving MM); once processed, it travels back via the data bus to the OU, which returns the result.

Answer 15. Performance characteristics:

- Processor: clock frequency (GHz), number of cores, cache-memory size.
- Memory (RAM): capacity (GB), frequency/speed (MHz), access time.
- Storage units: capacity (GB/TB), read/write speed, technology used (HDD hard disk or SSD).

Answer 16. Algorithm (pseudo-algorithm in natural language) for going home:

1. Walk to the entrance of the building.
2. Open the building entrance door.
3. Go to the elevator.

4. Press the elevator call button.
5. Wait for the elevator to arrive.
6. As soon as the door opens, enter the elevator.
7. Press the button corresponding to your floor.
8. Wait for the elevator to reach the desired floor.
9. Exit the elevator.
10. Go to the door of your apartment.
11. Open the door with the key.
12. Enter the apartment.

Answer 17. Algorithm for solving the equation $ax^2 + bx + c = 0$ in R:

```
Algorithm SolveQuadratic
Data: a, b, c: real numbers (with a != 0)
Result: the real solution or solutions of the equation
Begin
    Read(a, b, c)
    If (b*b - 4*a*c) > 0 then
        x1 <- (-b + square_root(b*b - 4*a*c)) / (2*a)
        x2 <- (-b - square_root(b*b - 4*a*c)) / (2*a)
        Display(x1, x2)
    Else If (b*b - 4*a*c) = 0 then
        x1 <- -b / (2*a)
        Display(x1)
    Else
        Display("No solution in R")
    EndIf
End
```

In mathematical notation: if $b^2 - 4ac > 0$, then $x_1 = (-b + \sqrt{(b^2 - 4ac)}) / (2a)$ and $x_2 = (-b - \sqrt{(b^2 - 4ac)}) / (2a)$; if $b^2 - 4ac = 0$, then $x_1 = x_2 = -b / (2a)$; if $b^2 - 4ac < 0$, there is no solution in R.

Answer Key for the Additional Questions (18 to 30)

Answer 18. Conversion of 2.5 GB:

- 2.5 GB = 2.5 × 1024 MB = 2560 MB
- 2560 MB = 2560 × 1024 KB = 2,621,440 KB
- 2,621,440 KB = 2,621,440 × 1024 bytes = 2,684,354,560 bytes

Answer 19.

97 in decimal corresponds to 01100001 in binary (64 + 32 + 1 = 97), so ASCII('a') = $(97)_{10} = (01100001)_2$.

Answer 20.

The data bus is a set of electrical lines (wires) that serves as a communication channel for transporting data between the different components of the computer: between the input unit and main memory, between main memory and the ALU (under the control of the CU), and between main memory and the output unit.

Answer 21.

A computer program is a sequence of instructions and data intended to be executed by a computer, generally performing a specific task. Software is a set of programs intended to carry out a number of tasks and services that meet specific needs; software can therefore group together several programs.

Answer 22. The three types of software:

- Operating system: manages the computer's resources and acts as an intermediary between the user and the machine. Example: Windows, Linux.
- Standard software: meets common needs of a broad public. Example: MS Word (word processing), MS Excel (spreadsheet).
- Software packages: specific software developed to meet a particular business need. Example: staff payroll software, inventory-management software.

Answer 23.

The word "Algorithm" comes from the name of the Persian mathematician Al-Khwarizmi (in Arabic: الخوارزمي), distorted in Ancient Greek into "arithmós," a word meaning "number."

Answer 24. The five properties of an algorithm:

- Generality: the algorithm must consider all possible cases of the processing, not just one particular case.
- Finiteness: the algorithm must stop after a finite amount of time.
- Definiteness: all operations must be defined without ambiguity.
- Repetitiveness: an algorithm generally contains several iterations, i.e. actions that are repeated.
- Efficiency: the algorithm should run in minimal time and consume a minimum of resources.

Answer 25. Property not respected in each case:

- a) Finiteness (the algorithm does not stop).
- b) Definiteness (the instruction is ambiguous: "approximately" and "small number" are not defined precisely).
- c) Generality (the algorithm handles only one particular case instead of all possibilities).
- d) Efficiency (the algorithm works but is not optimized in time/resources).

Answer 26.

A pseudo-algorithm is an algorithm expressed in natural language (and not in a formal algorithmic language), describing a sequence of actions that makes it possible to perform a task. Example: instructions for sending a fax - "Insert the document into the automatic feeder, dial the recipient's fax number, press the send button."

Answer 27. Algorithm for calculating the sum of two integers:

```
Algorithm SumOfTwoNumbers
Data: a, b: integers
Result: s: integer
Begin
    Read(a)
    Read(b)
    s <- a + b
    Display(s)
End
```

Answer 28. Algorithm for the largest of three real numbers:

```
Algorithm LargestOfThree
Data: a, b, c: real numbers
Result: max: real number
Begin
```

```
    Read(a, b, c)
    max <- a
    If b > max then
        max <- b
    EndIf
    If c > max then
        max <- c
    EndIf
    Display(max)
End
```

Answer 29. Pseudo-algorithm (natural language) for preparing a cup of coffee/tea:

13. Fill the kettle with water.
14. Heat the water until it boils.
15. Put one spoonful of coffee (or a tea bag) in the cup.
16. Pour the hot water into the cup.
17. Add sugar to taste.
18. Stir with a spoon.
19. Serve.

Answer 30. Capacity calculation:

- 4 GB = 4 × 1024 MB = 4096 MB
- 4096 MB = 4096 × 1024 KB = 4,194,304 KB
- 4,194,304 KB = 4,194,304 × 1024 bytes = 4,294,967,296 bytes

Since each ASCII character occupies exactly 1 byte (8 bits), a maximum of 4,294,967,296 characters can therefore be stored, i.e. approximately 4.29 billion characters (which corresponds exactly to 2^{32} bytes).